

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Easy

Question

While researching a topic, a student has taken the following notes:

- Some atoms contain an excess of neutrons.
- Often, these neutrons form a “skin” on the atom’s surface.
- An atom of lead-208 has a neutron skin.
- The thickness of its neutron skin is approximately 0.28 trillionths of a millimeter.

The student wants to emphasize the thickness of lead-208’s neutron skin. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. The neutron skin surrounding an atom of lead-208 measures about 0.28 trillionths of a millimeter.
- B. Atoms with excess neutrons will often acquire a neutron skin.
- C. An atom of lead-208, like some other atoms, is surrounded by a neutron skin.
- D. Neutrons surround the surface of an atom of lead-208.

Correct Answer: A

Rationale

Choice A is the best answer. The sentence emphasizes the thickness of lead-208’s neutron skin, noting that it is about 0.28 trillionths of a millimeter thick.

Choice B is incorrect. The sentence makes a generalization about atoms, stating that atoms with excess neutrons will often acquire a neutron skin; it doesn’t emphasize the thickness of lead-208’s neutron skin. Choice C is incorrect. The sentence states that lead-208 has a neutron skin; it doesn’t emphasize the thickness of that skin. Choice D is incorrect. The sentence indicates that lead-208 is surrounded by neutrons; it doesn’t emphasize the thickness of lead-208’s neutron skin.